

STUTTGART ECHTERDINGEN, Germany

WMO: 107380

Lat: 48.6883N

Lon: 9.2236E

Elev: 1284

StdP: 14.03

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.0	19.1	7.0	8.2	16.2	12.6	10.9	20.8	24.3	43.8	21.8	43.7	4.7	40	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	88.1	67.2	84.3	66.2	80.9	65.3	69.4	82.5	67.9	80.5	66.7	78.5	6.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	96.2	73.9	63.0	90.5	72.1	62.5	88.6	71.5	34.2	82.1	33.1	80.8	32.0	78.0	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.0	13.8	DB	9.4	94.2	5.0	2.2	5.8	95.8	2.9	97.1	0.1	98.3	-3.5	99.9	(4)
(5)			WB	8.6	72.3	5.0	1.7	5.0	73.5	2.1	74.5	-0.7	75.5	-4.3	76.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	34.2	35.2	42.4	51.1	57.2	64.6	67.7	66.0	59.5	51.3	42.3	36.7	(6)	
	DBStd	13.69	7.94	7.89	7.65	6.93	6.61	6.44	5.87	5.95	5.71	6.76	6.73	7.31	(7)	
	HDD50	1969	493	414	249	70	14	1	0	0	2	68	244	414	(8)	
	HDD65	5511	956	833	701	417	255	86	40	59	182	424	680	878	(9)	
	CDD50	2252	2	1	13	103	236	439	550	497	287	109	14	1	(10)	
	CDD65	319	0	0	0	0	12	74	125	91	17	0	0	0	(11)	
	CDH74	3155	0	0	0	32	160	718	1211	855	170	9	0	0	(12)	
	CDH80	969	0	0	0	2	31	204	425	279	28	0	0	0	(13)	
(14) Wind	WSAvg	5.5	6.2	6.1	6.6	5.8	5.9	5.4	5.3	4.8	4.6	4.6	5.1	5.7	(14)	
(15) Precipitation	PrecAvg	28.3	1.7	1.5	1.8	2.1	3.1	3.7	3.0	3.0	2.2	1.9	2.2	1.8	(15)	
	PrecMax	43.7	4.8	4.8	6.2	5.3	6.6	7.7	7.0	7.9	5.7	5.9	4.9	4.5	(16)	
	PrecMin	18.1	0.2	0.3	0.3	0.4	1.1	0.6	0.3	0.7	0.6	0.1	0.4	0.1	(17)	
	PrecStd	5.4	1.0	0.9	1.1	1.2	1.4	1.7	1.4	1.6	1.1	1.3	1.1	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	59.2	66.6	77.3	84.2	91.0	93.4	91.7	83.9	75.1	64.6	56.8	(19)	
		MCWB	49.1	47.5	51.4	57.7	63.9	68.5	68.6	67.4	64.4	60.7	55.9	50.3	(20)	
	2%	DB	52.2	53.3	62.8	73.6	78.7	85.7	89.2	87.4	78.8	69.7	59.2	53.2	(21)	
		MCWB	47.4	45.9	49.3	55.3	62.3	66.9	67.0	67.4	63.8	59.3	52.6	48.6	(22)	
	5%	DB	48.5	49.6	58.8	69.8	73.7	81.6	84.5	82.5	75.1	66.1	55.7	49.9	(23)	
		MCWB	44.6	44.3	48.3	54.1	59.0	65.6	66.3	66.7	63.0	57.9	50.9	46.1	(24)	
	10%	DB	44.9	46.3	55.1	64.8	70.0	77.6	80.7	78.7	71.4	62.5	53.2	46.6	(25)	
		MCWB	42.3	42.0	46.9	52.2	58.2	64.2	65.4	65.3	61.1	56.1	49.1	43.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	49.3	53.4	59.0	66.9	71.4	71.1	70.9	67.2	62.3	56.5	51.5	(27)	
		MCDB	54.8	56.2	62.6	74.2	79.0	84.7	85.5	83.2	77.7	70.7	62.9	54.8	(28)	
	2%	WB	48.3	46.6	50.9	57.0	63.9	68.6	69.1	69.1	65.3	60.4	53.4	49.1	(29)	
		MCDB	52.8	52.6	59.0	70.5	75.1	80.9	82.8	81.7	75.6	67.5	57.9	52.6	(30)	
	5%	WB	44.9	44.2	49.3	54.9	61.4	67.0	67.7	67.6	63.6	58.5	51.2	46.4	(31)	
		MCDB	48.1	48.6	56.7	66.9	70.7	78.7	80.5	79.9	73.5	64.3	55.7	50.0	(32)	
	10%	WB	42.3	41.7	47.3	53.1	59.1	65.0	66.4	65.9	61.7	56.8	49.1	43.7	(33)	
		MCDB	44.7	45.4	54.5	63.8	68.2	76.2	78.3	76.9	69.6	61.9	52.9	46.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	11.9	16.3	18.9	17.6	18.6	19.0	18.5	17.9	16.2	12.0	10.7	(35)	
		MCDBR	13.9	19.1	25.2	28.2	25.4	25.8	26.6	25.8	24.1	22.9	17.5	14.5	(36)	
	5% WB	MCWBR	10.7	12.7	14.3	13.4	11.9	11.3	9.7	9.9	11.8	13.5	12.3	10.9	(37)	
		MCWBR	13.4	17.1	21.5	24.6	22.0	23.4	22.8	23.5	21.8	19.8	16.2	13.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.329	0.360	0.394	0.404	0.409	0.409	0.405	0.378	0.353	0.325	0.301	(40)	
		taud	2.446	2.387	2.331	2.264	2.266	2.286	2.278	2.309	2.364	2.427	2.460	2.449	(41)	
	Ebn at Noon		243	260	270	271	272	271	269	264	260	248	232	228	(42)	
		Edh at Noon	20	27	33	39	41	41	40	37	32	26	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		301	544	920	1347	1590	1813	1700	1484	1097	666	349	240	(44)	
	RadStd		41	89	134	190	171	145	153	143	120	70	43	40	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	+0.65	+0.59	N/A	N/A	N/A	+40

Nomenclature: See separate page